A Multimodal Neuroimaging Case Study of Blindsense in Cortical Blindness

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Background

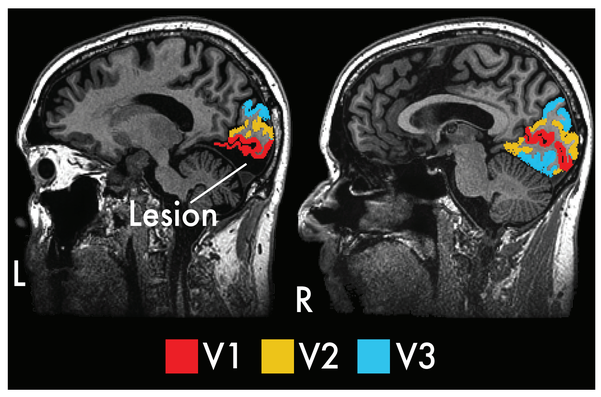
- Cerebral blindness is the complete or partial loss of vision following damage to the visual pathway (e.g., V1). [1][2]
- Some people with cortical blindness experience impaired or residual vision in their blind field, referred to as blindsense. [3]
- P4 is a cerebrally blind person (right homonymous superior quadrantanopia), caused by a stroke in the left visual cortex.
- P4 experiences non-visual "sensations" or "feelings" for some visual stimuli presented in the blind field. [4]

Main Question

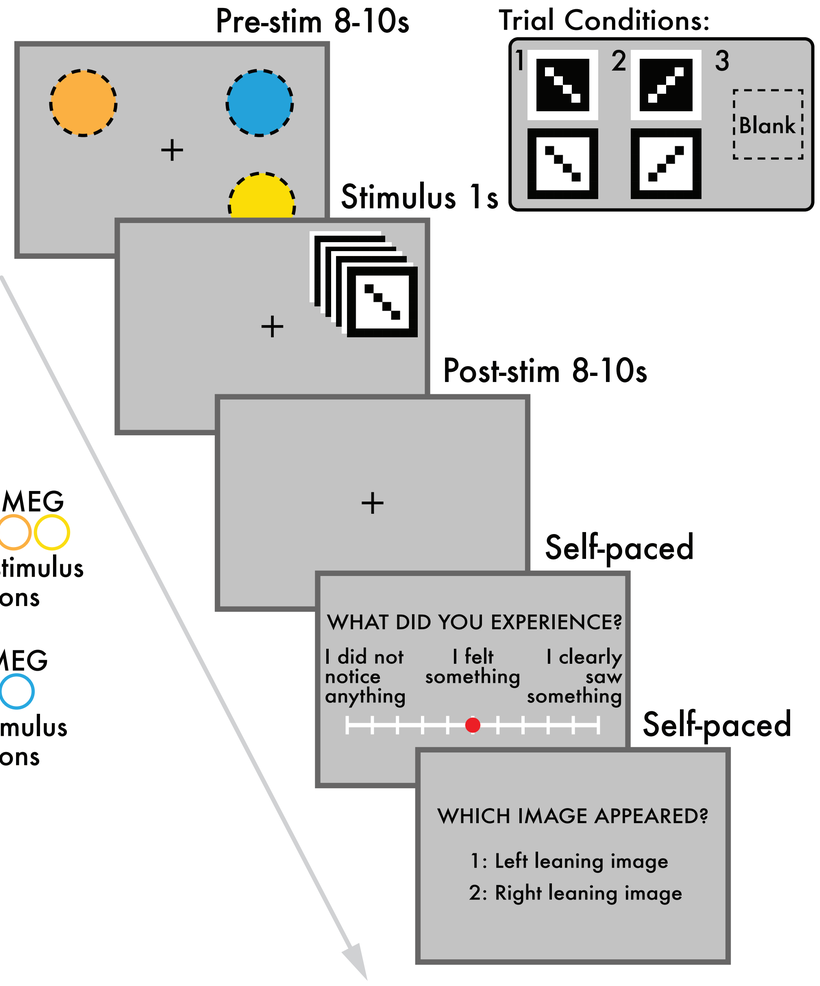
What are the brain mechanisms of residual vision/blindsense in cortical blindness?

Methods

Lesion & Blind Field

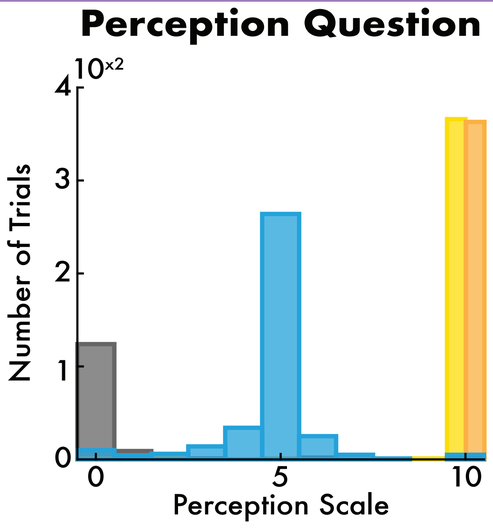


Paradigm

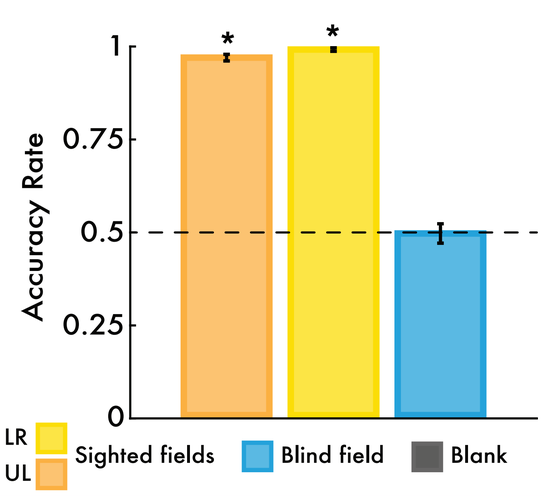


P4 reported "I felt something" for most stimuli shown in the blind field; chance orientation discrimination

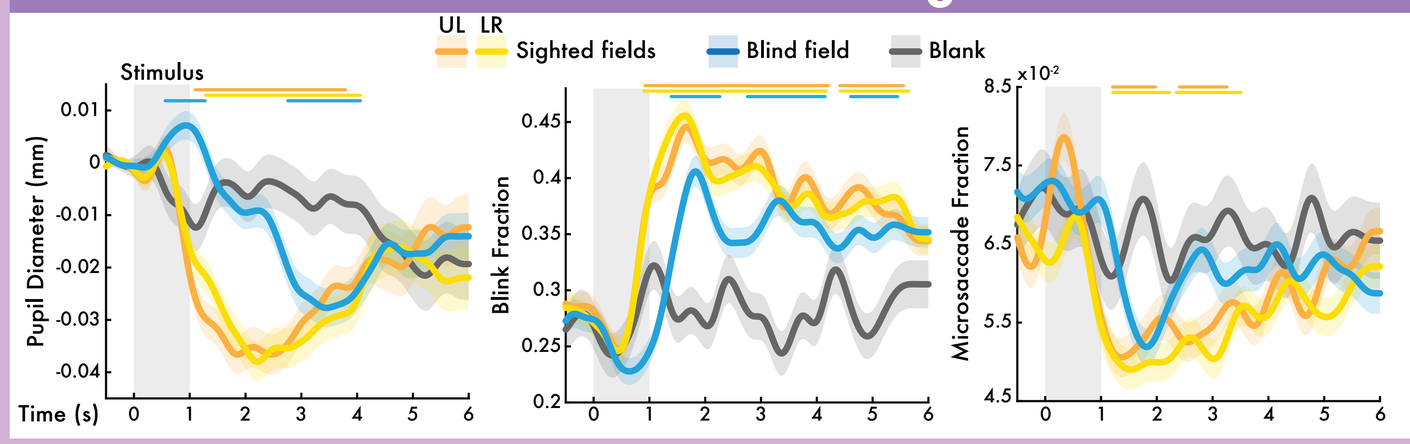
Perception Question



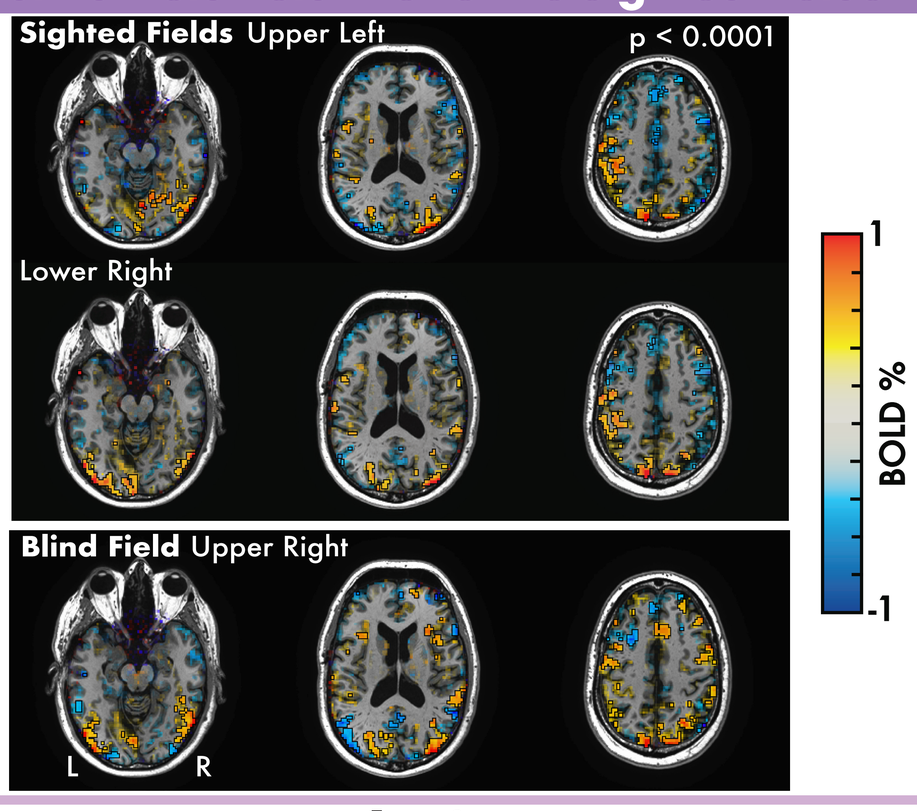
Orientation Question



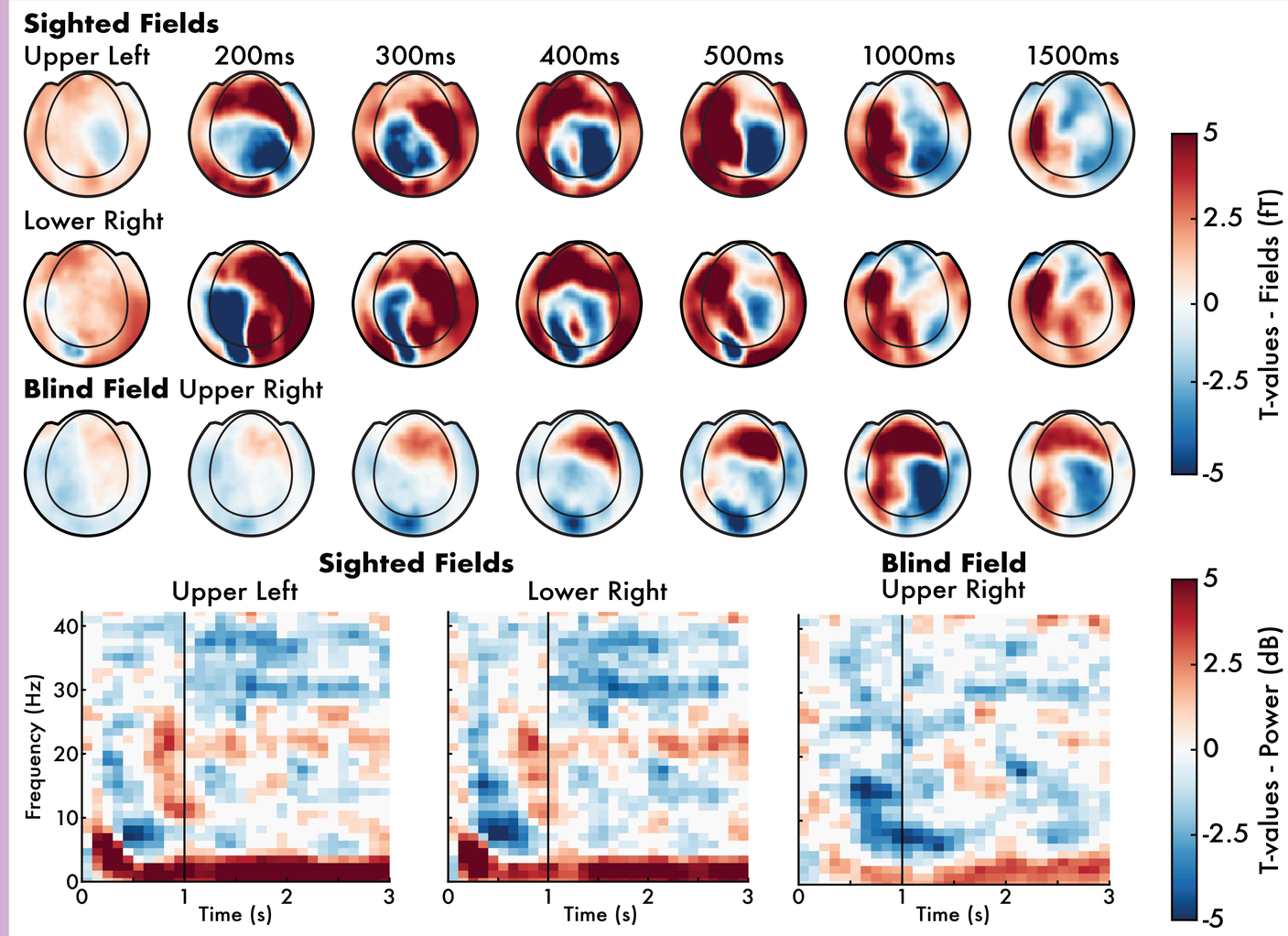
Eye behavior shows decreased and delayed responses in the blind field relative to the sighted fields



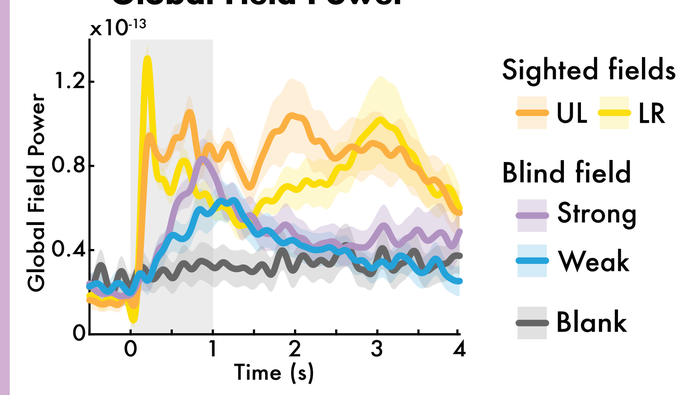
Convergent and divergent BOLD activity in the blind field relative to the sighted fields



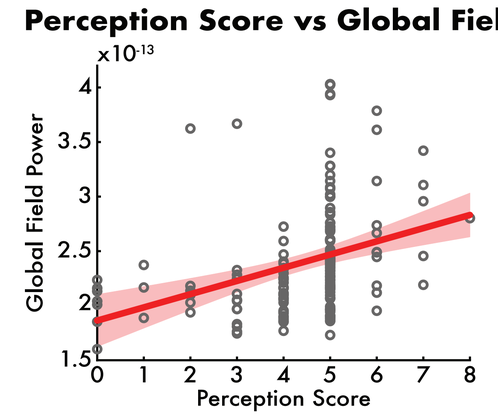
Decreased and delayed fields; distinct time-frequency patterns in the blind field relative to the sighted fields



Global Field Power



Perception Score vs Global Field Power



Conclusions

- Eye behavior showed decreased and delayed responses in the blind field relative to the sighted fields.
- MEG fields showed decreased and delayed responses in the blind field, with signal power positively correlated with perception score.
- fMRI activity shows that the blind field stimuli engage convergent visual cortex regions but distinct salience and frontal network regions.

Acknowledgements
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